AE646 - Individual Contribution Statement & AI-Tool-Use Declaration

Team PINNacles, Stage 3 (final)

Project: Optimising a Fourier Neural Operator for parametric Darcy flow (FNO vs. MLP baseline, comparison with a finite-volume solver), on real PDEBench data.

Member	Contribution
Aryamann Srivastava	Lead for implementation and experiments. Problem formulation and literature review; data pipeline (download, preprocessing, 900/100/200 split); MLP and FNO models; Stage 2 training, evaluation, profiling, EDA and ablation scripts; Stage 3 experiment framework (`stage3_train.py`, job queue, D4 augmentation and test-time symmetrisation), recipe/architecture/data/resolution studies and the five-seed final models, finite-volume solver with calibration and timing, latency benchmark, physical diagnostics, aggregation and figures; ran the Stage 3 experiments on GPU runtimes (reproduced on Google Colab and Kaggle, T4 GPU); wrote the Stage 1–3 reports and presentations and the code package.
Varun Sathaye	Data and baseline verification: checking the train/validation/test split and normalisation, reproducing the Stage 2 baseline results. Stage 3: solver-vs-FNO comparison and latency (report §7): reviewing the calibration and timing results, and presenting them.
Atishay Jain	Stage 3 training-recipe and architecture studies (report §4–§5): reviewing the ablation and sweep results, the model-selection procedure and the final-model tables, and presenting them; review of report text and figures.
Vedant S. Tiwari	Stage 3 data-scaling, resolution and physical-interpretation studies (report §6, §8): reviewing the diagnostics and failure-case analysis, and presenting them; literature and background check (FNO, PDEBench).

Code and results: The code.zip file contains our codes for each particular task. Each member can walk through the parts listed against their name in the viva.

AI-tool-use declaration. In line with the AE646 policy, AI tools were used (i) for coding and debugging of the project scripts, including the Stage 3 experiment framework, solver, benchmarks and diagnostics; (ii) at our direction, to run the experiments (Google Colab and Kaggle, T4 GPU runtimes) and cross-check every number in the reports, slides and README against the stored result files, which uncovered the suboptimal Stage 2 training recipe and the invalid Stage 2 solver-timing reference (both reported in the final report); and (iii) for LaTeX and `python-pptx` drafting and formatting. The experimental design, results and interpretation were reviewed by the team. All numerical results come from real runs on real data; none were generated or invented by an AI tool.

Date: November 2026